



SongCluster

A model for automatic Spotify playlist generation



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NG

Introducing our Model

SongCluster

- Prototype model
- Creates Spotify playlists based on mood
- Quick, efficient categorisation of large numbers of songs

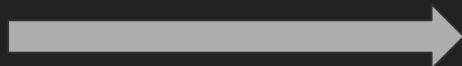


Here, we present our model and some of the playlists it has created

SongCluster

We tested our model on a Spotify dataset

- 5171 songs
- 2176 artists



How does it work?

- Clustering using unsupervised machine learning



Our Data

The playlists are based on song features provided directly by Spotify.

Included Features:

tempo 

energy 

danceability 

Excluded Features:

artist name 

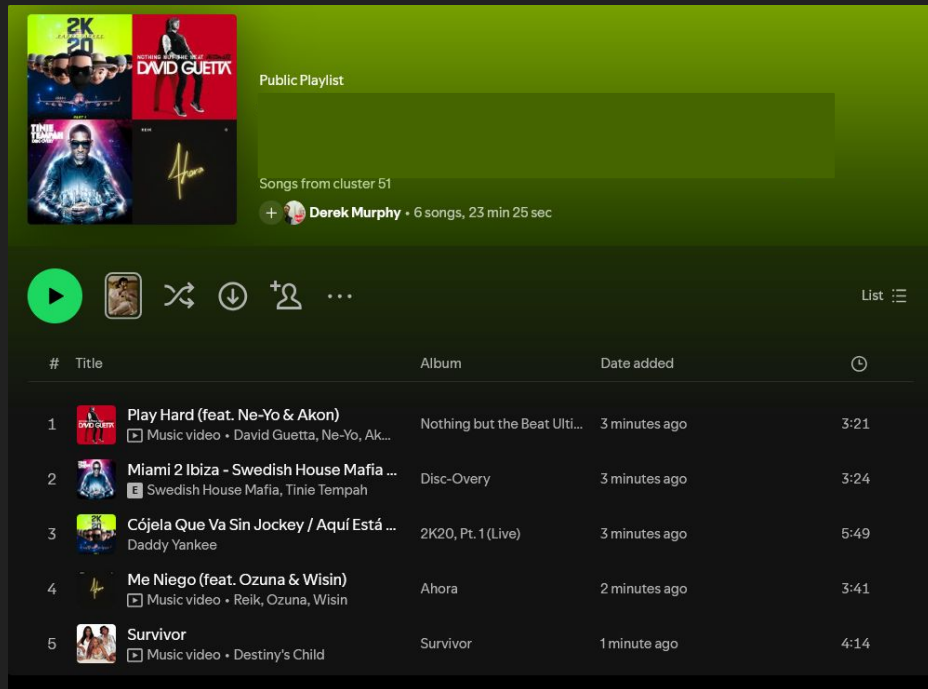
key 

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Exploring the playlists

Playlist 51

- Danceable, high energy, loud, upbeat songs
- Hip hop, reggaeton, electronica, eurodance
- 'Dance party!' 🕺 🕺 🕺 🕺



Public Playlist

Songs from cluster 51

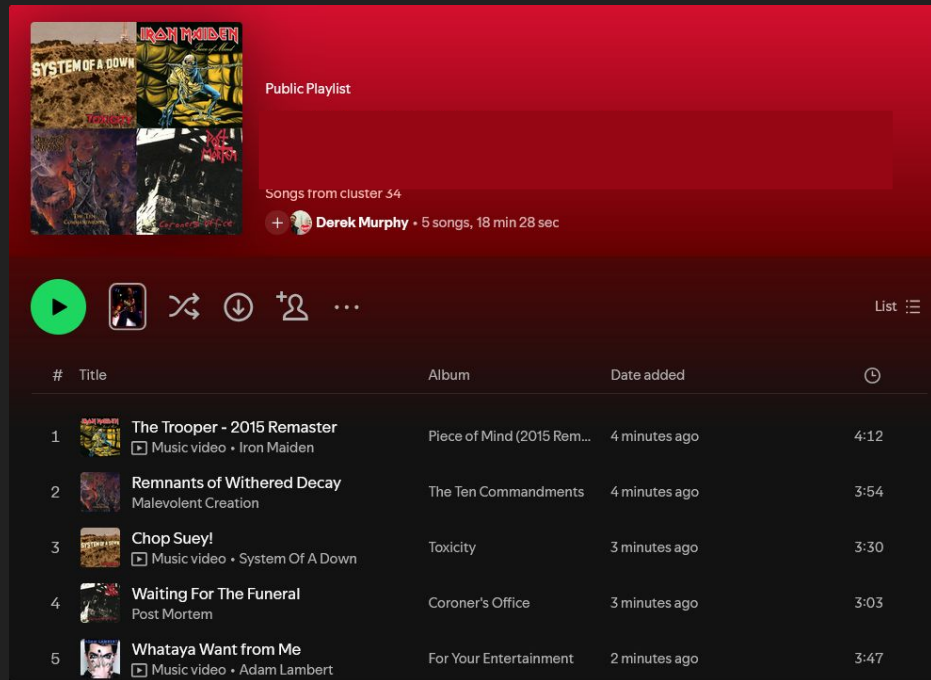
+ Derek Murphy • 6 songs, 23 min 25 sec

Play icon, Shuffle, Repeat, Download, Add to playlist, More options

#	Title	Album	Date added	Duration
1	 Play Hard (feat. Ne-Yo & Akon) Music video • David Guetta, Ne-Yo, Akon...	Nothing but the Beat Ulti...	3 minutes ago	3:21
2	 Miami 2 Ibiza - Swedish House Mafia ... Swedish House Mafia, Tinie Tempah	Disc-Overy	3 minutes ago	3:24
3	 Cójela Que Va Sin Jockey / Aquí Está ... Daddy Yankee	2K20, Pt. 1 (Live)	3 minutes ago	5:49
4	 Me Niego (feat. Ozuna & Wisin) Music video • Reik, Ozuna, Wisin	Ahora	2 minutes ago	3:41
5	 Survivor Music video • Destiny's Child	Survivor	1 minute ago	4:14



Exploring the playlists



Public Playlist

Songs from cluster 34

+ Derek Murphy • 5 songs, 18 min 28 sec

#	Title	Album	Date added	
1	 The Trooper - 2015 Remaster Music video • Iron Maiden	Piece of Mind (2015 Rem...	4 minutes ago	4:12
2	 Remnants of Withered Decay Malevolent Creation	The Ten Commandments	4 minutes ago	3:54
3	 Chop Suey! Music video • System Of A Down	Toxicity	3 minutes ago	3:30
4	 Waiting For The Funeral Post Mortem	Coroner's Office	3 minutes ago	3:03
5	 Whataya Want from Me Music video • Adam Lambert	For Your Entertainment	2 minutes ago	3:47

Playlist 34

- High energy, very loud, dark songs with low danceability
- Death metal, Nu-Metal, Alternative, Pop-Rock
- ‘Headbangers!’ 🦴 🎸 🤘

Exploring the playlists

Playlist 7

- slow, instrumental, low energy, quiet
- Ambient, classical, instrumental
- 'Calm & Meditative!' 🧘 🙏 🙌 😊

Public Playlist

Songs from cluster 7

+ Derek Murphy • 5 songs, 37 min 22 sec

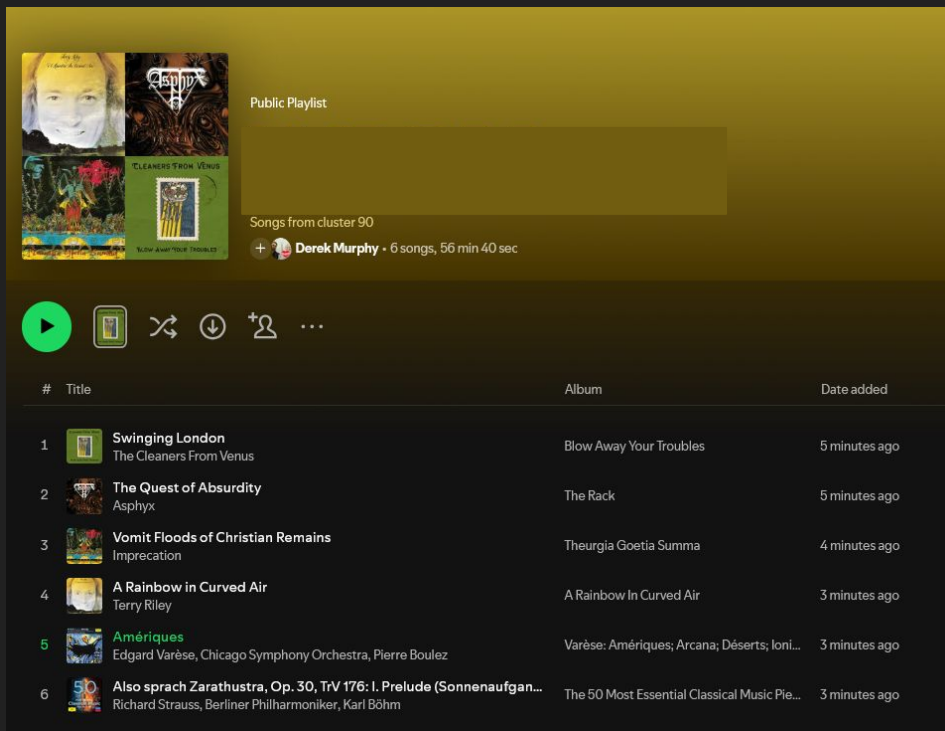
Play button, Shuffle, Repeat, Download, Add to library, More options

List

#	Title	Album	Date added	🕒
1	Space 11 (Invisible Pages Over) Max Richter, Ben Russell, Yuki Numata Res...	From Sleep	31 seconds ago	5:15
2	Water Memory 1 Emily A. Sprague	Water Memory	24 seconds ago	7:28
3	Beethoven 250 A Winged Victory for the Sullen	@0 EP2	16 seconds ago	4:11
4	Symphony No. 3 in A Minor, Op. 44: I... Sergei Rachmaninoff, Philadelphia Orches...	Rachmaninoff: Symphoni...	6 seconds ago	13:46
5	Stave Peak Loscil	Coast / Range / Arc	1 second ago	6:40



Exploring the playlists



Public Playlist

Songs from cluster 90

+ Derek Murphy · 6 songs, 56 min 40 sec

#	Title	Album	Date added
1	 Swinging London The Cleaners From Venus	Blow Away Your Troubles	5 minutes ago
2	 The Quest of Absurdity Asphyx	The Rack	5 minutes ago
3	 Vomit Floods of Christian Remains Imprecation	Theurgia Goetia Summa	4 minutes ago
4	 A Rainbow in Curved Air Terry Riley	A Rainbow in Curved Air	3 minutes ago
5	 Amériques Edgard Varèse, Chicago Symphony Orchestra, Pierre Boulez	Varèse: Amériques; Arcana; Déserts; Ioni...	3 minutes ago
6	 Also sprach Zarathustra, Op. 30, TrV 176: I. Prelude (Sonnenaufgan... Richard Strauss, Berliner Philharmoniker, Karl Böhm	The 50 Most Essential Classical Music Pie...	3 minutes ago

It doesn't always work well

Playlist 90

- Mostly dark and moody but high variability in energy, tempo
- Ambient, Death Metal, Classical, Indie
- 'Messy' 💥🤔

Additional Signals for Better Playlist Generation



Genre

- Example: Alternative Metal → “Chop Suey”



User Listening Behaviour

- Example: Reggaeton fans → “Play Hard”



Lyrics / Mood

- Example: Calm / melancholic → “Beethoven 250”



Artist's Style

- Example: Electronic artists → “David Guetta”



Audio Features + Human Perception  **Better Playlist Generation**

Why SongCluster?

Pros

- Fast and scalable
- Compatible with data from Spotify
- Produces interpretable song groupings
- Flexible number of playlists

Cons

- Difficult to ensure a perfect playlist fit
- Cannot control playlist size
- Requires data pre-processing

Next steps?

- Expand the dataset to the entire Spotify database
- More features → better classification
- Label playlists by mood
- Human-curated playlists with ML support